

JIARUI HE

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EDUCATION

National University of Singapore

MSc Building Performance & Sustainability

Singapore
2025.08 – Present

- GPA: 4.33/5 (86.6)
- Dissertation: BIM-IoT Enabled Digital Twin for Real-Time Analytics and Predictive Control of Building Energy Systems.
- Core: Passive/Active System, Building integrated diagnosis, Data-driven Decision Making for Smart Building, Smart Buildings and Facilities, etc.
- Mainly research work on: Digital Twin, Advanced Control (Reinforcement learning and MPC), BIM, Building Simulation, LLM(RAG, transformer), Fault Detection and Diagnosis, Machine learning and Building Energy Flexibility.

Hong Kong Polytechnic University

BSc (HONS) Building Engineering & Management

Hong Kong, China
2021.08 – 2025.07

- GPA: 3.42/4.3 (80) / Second Class Honours (Upper Division)
- Core: Building Science I-II, Building Information and Modelling, Data Algorithm and Structure, Building advanced material and maintenance, Analytical Skills and Methods, Engineering Mathematics etc.

RESEARCH EXPERIENCES

National University of Singapore, Building Informatics & Operation Research (BIOR) Lab

MSc Student

Singapore
2025.08 – Present

Supervisor: Dr. Hu Maomao

- Developing an Energy Management and Information System (EMIS) for smart buildings, including data analytics, visualization toolkit, predictive control, and fault detection and diagnosis (FDD)
- Building a BIM-IoT-enabled digital twin platform using Unity and web-based interfaces for real-time monitoring and decision support
- Designing data pipelines for real-time and historical building data collection, storage, and analysis using Linux servers, InfluxDB, FastAPI, and MQTT

Hong Kong Polytechnic University

Research Assistant, Project "Automation Building Assessment through Scan-to-BIM"

Hong Kong, China
2024.05 – 2025.03

Supervisor: Prof. Albert P. C. Chan

- Supported the development of BIM models for interior space assessment and the integration of IoT-related building information
- Assisted in reviewing and analyzing green building certification frameworks, including BEAM Plus benchmarks
- Co-authored academic papers for journal publications

AWARDS

2024 Fund for Innovative Technology-in-Education projects (FITE)

- Dissertation Funding Grant, selected by the Dept. of Building and Real Estate (BRE), PolyU.

2022 – 2023 Hong Kong Polytechnic University - Micro Fund for College Students Entrepreneurship

The Hong Kong Polytechnic University (PolyU) Finalist "Energy Group"

- Participated in the 5-week Lean Launchpad Program for Market Validation from HKSTP, shortlisted for the final startup panel and got a \$100,000 HKD startup funding slot.

2022 Qianhai-Guangdong-Hong Kong-Macao-Taiwan Youth Innovation and Entrepreneurship Competition

Special 1st Golden Award | Technology & Sustainability Track (Prof. Ni Meng)

- Core team member of "Zinc Air Battery", a sustainable storage energy project as technician consultant for the battery material performance and property.

SKILLS

Languages: Native English and Mandarin | German (B2 level)

Software & Platforms:

Revit, Rhino Grasshopper, Unity, Unreal Engine, DesignBuilder, Ansys Fluent CFD, Grafana, Microsoft Office

Programming & Data Tools:

Python, JavaScript, C++, R, FastAPI, InfluxDB, Docker, MQTT, Linux (Ubuntu)